

Thi-Nhung Nguyen

PhD Student | Agentic AI & Frontier Models

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[Personal Website](#) | [Google Scholar](#)

PROFILE

PhD student with 5+ years in machine learning research and production AI. *My research focuses on safety, autonomy, and robustness in a Society of Intelligent Agents.* I aim to advance agentic systems and frontier models and enable large-scale coordination through agentic collectives and economies. My broader work spans intelligent orchestration and communication, self-improvement, multi-agent coordination, LLM alignment, long-horizon planning and reasoning, retrieval-augmented generation, and model evaluation. My work has been published at EMNLP, COLM, CoNLL, NAACL, and LREC-COLING.

Before my PhD, I led LLM and RAG products from experimentation to deployment at [MISA](#) and conducted research on weakly supervised learning and information extraction as an AI Research Resident at VinAI Research (now part of [Qualcomm](#)).

EDUCATION

Monash University

Doctor of Philosophy, Data Science & AI

Jun. 2025–Present

Melbourne, Australia

- Research: Safe and Autonomous Multi-Agent Systems.
- Supervisors: [Prof. Dinh Phung](#), [Dr. Thuy-Trang Vu](#), and [Dr. Linhao Luo](#).

Hanoi University of Science and Technology

Bachelor of Computer Science

Aug. 2017–Aug. 2021

Hanoi, Vietnam

- Thesis: *An Uncertainty-Aware Encoder for Aspect Detection*; Best Thesis Presentation Award.

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch, Hugging Face Transformers, FastAPI, Rasa, LangChain, LangGraph.

LLMs, Post-Training & Evaluation: Fine-tuning, alignment, reinforcement learning, automated data generation, model evaluation.

RAG & Retrieval: RAG, hybrid search, Text-to-SQL, Text-to-Cypher.

Agentic AI: AI agents, multi-agent systems, planning, workflow orchestration, safety and evaluation.

Databases & Infrastructure: Elasticsearch, Qdrant, Neo4j, MongoDB, Docker.

RESEARCH EXPERIENCE

PhD Researcher & Research Assistant

Monash University

Jun. 2025–Present

Melbourne, Australia

- Supervisors: [Prof. Dinh Phung](#), [Prof. Thuy-Trang Vu](#), and [Dr. Linhao Luo](#).
- Conduct research on safe, autonomous, and robust multi-agent systems, encompassing LLM alignment, long-horizon planning and reasoning, intelligent orchestration and communication, self-improvement, and safety evaluation.
- Developed research and evaluation methods for truth consistency, long-term planning, retrieval-augmented generation (RAG), table understanding, and emergent bias in multi-agent systems.
- Contribute to a multi-agent safety project (Jan. 2026–Present) and contributed to an LLM alignment project (Apr.–Jun. 2026).
- Published three papers at EMNLP, COLM, and CoNLL, with additional work under review.

AI Research Resident

VinAI Research (now part of Qualcomm AI Research)

Mar. 2022–Apr. 2024

Hanoi, Vietnam

- Research supervisors: [Assoc. Prof. Thien Huu Nguyen](#) and [Dr. Dat Quoc Nguyen](#).
- Researched weakly supervised learning and information extraction, resulting in four papers at EMNLP, NAACL, and LREC-COLING.
- **PhoGPT**: contributed to an open-source generative pre-trained language model for Vietnamese.

Research Student

Data Science Laboratory, HUST

Oct. 2019–Mar. 2022

Hanoi, Vietnam

- Supervisor: [Dr. Kiem-Hieu Nguyen](#).
- Developed weakly supervised methods for aspect detection; published at EMNLP Findings 2021.

INDUSTRY EXPERIENCE

Senior AI Engineer, LLM Team Lead <i>MISA JSC</i> - Supervisor: Dr. Trung Bui (Adobe Research). - Led fine-tuning and evaluation of Vietnamese LLMs, reaching #1 on VMLU, a Vietnamese-language LLM benchmark, as of Mar. 25, 2025 and a top-five rank in a national government evaluation of Vietnamese LLMs. - Built automated data-generation and evaluation pipelines, fine-tuned embedding models, local LLMs, and domain agents for accounting and financial analysis. - Directed delivery across LLM alignment and safety, RAG, and multi-agent projects while mentoring engineers and communicating with stakeholders.	Dec. 2024–May 2025 <i>Hanoi, Vietnam</i>
Senior AI Engineer, Technical Lead <i>MISA JSC – Legal Q&A System</i> - Led the development of a RAG-based legal assistant covering 1,000+ documents (3–300 pages each), achieving > 95% search accuracy and > 90% answer accuracy on 1,000+ expert-curated questions. - Translated product requirements into retrieval and evaluation tasks using Python, Rasa, databases, chunking, RAG, and Text-to-SQL; product success helped secure two new projects.	Apr. 2024–Dec. 2024 <i>Hanoi, Vietnam</i>

SELECTED PUBLICATIONS

TriAlign: Towards Universal Truth Consistency in Personalized LLM Alignment Thi-Nhung Nguyen , Linhao Luo, Rollin Omari, Junae Kim, Thuy-Trang Vu, Dinh Phung <i>EMNLP 2026</i>	
The Social Cost of Intelligence: Emergence, Propagation, and Amplification of Stereotypical Bias in Multi-Agent Systems Thi-Nhung Nguyen , Linhao Luo, Amardeep Kaur, Rollin Omari, Tamas Abraham, Junae Kim, Thuy-Trang Vu, Dinh Phung <i>Preprint, 2026</i>	
Improving Table Understanding with LLMs and Entity-Oriented Search Thi-Nhung Nguyen , Hoang Ngo, Dinh Phung, Thuy-Trang Vu, Dat Quoc Nguyen <i>COLM 2025</i>	
Planning for Success: Exploring LLM Long-term Planning Capabilities in Table Understanding Thi-Nhung Nguyen , Hoang Ngo, Dinh Phung, Thuy-Trang Vu, Dat Quoc Nguyen <i>CoNLL 2025</i>	
BKEE: Pioneering Event Extraction in the Vietnamese Language Thi-Nhung Nguyen , Bang Tran, Trong-Nghia Luu, Kiem-Hieu Nguyen, Thien Huu Nguyen <i>LREC-COLING 2024</i>	
A Self-Enhancement Multitask for Unsupervised Aspect Category Detection Thi-Nhung Nguyen , Hoang Ngo, Kiem-Hieu Nguyen, Tuan-Dung Cao <i>EMNLP 2023</i>	
An Uncertainty-Aware Encoder for Aspect Detection Thi-Nhung Nguyen , Kiem-Hieu Nguyen, Young-In Song, Tuan-Dung Cao <i>Findings of EMNLP 2021</i>	

TEACHING

Teaching Assistant – Natural Language Processing, Monash University	Semester 1, 2026
Teaching Assistant – Natural Language Processing, VinUniversity	2024
Teaching Assistant – Web Mining and Graph Analytics, HUST	2021

SELECTED HONORS

Best Thesis Presentation Award , HUST, 2021
Microsoft YouthSpark Scholarship , 2017 and 2018